

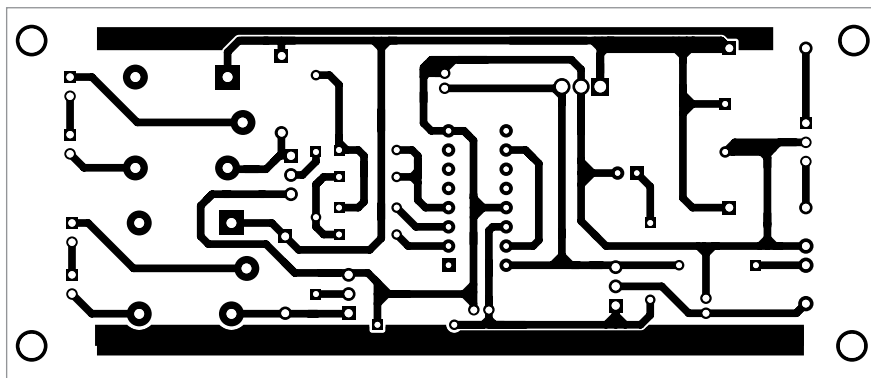


Fig. 3: Actual-size PCB layout for the remote control

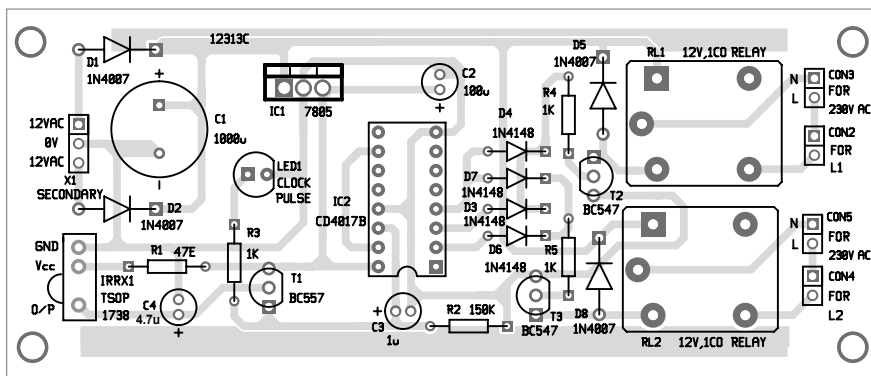


Fig. 4: Components layout of the PCB

circuit receives a train of 38kHz IR pulses. These pulses are inverted at the collector of T1 and given to the clock input of decade counter IC2.

The arriving pulses could increment the CD4017B counter at the same rate (38kHz) but, because of the RC filter circuit (comprising R2 and C3) at the clock input pin 14 of IC2, the train of pulses appear as a single pulse to the counter. Thus, on pressing each key, the CD4017B counter advances by a single count only.

When you release the remote's key, capacitor C3 discharges through resistor R2 and the clock line becomes zero. So, every time you press and releases a key on the remote, CD4017B counter receives a single pulse at its clock input. LED1 glows to indicate that a pulse has been received by IC2 from the remote key.

There can be five cases (types) of operation:

Case 1. Initially, when you press a key on the IR remote, the O_0 output of CD4017B goes high. The counter increments for each low-to-high going pulse arriving at its CLK pin (14). When the first pulse arrives, O_0

goes high and transistor T2 conducts through diode D3. This activates relay RL1 and appliance L1 connected to it is turned on.

Case 2. When you press a key again, the second pulse arriving at CLK line increments the counter by one. So, O_1 output of CD4017B goes high and transistor T3 conducts through diode D5. This activates relay RL2 and AC appliance L2 connected to it is turned on. At the same time, since output pin O_0 is low, transistor T2 is cut-off by diode D3. This deactivates relay RL1 and appliance L1 is turned off.

Case 3. When you press a key once again, the third pulse arriving at CLK line again increments the counter by one. So, O_2 output of CD4017B goes high and transistors T2 and T3 conduct through diodes D4 and D6, respectively. This activates both the relays (RL1 and RL2) and both the appliances (L1 and L2) are turned on.

Case 4. When you press a key for the fourth time, the fourth pulse arriving at CLK line increments the counter by one once more. So, O_3 output of CD4017B goes high and output pins O_0 , O_1 , and O_2 of CD4017B become

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - CD4017B decade counter
- D1, D2, D5, D8 - 1N4007 rectifier diode
- D3, D4, D6, D7 - 1N4148 signal diode
- LED1 - 5mm LED
- T1 - BC557 PNP transistor
- T2, T3 - BC547 NPN transistor
- IRRX1 - TSOP1738 IR receiver

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 47-ohm
- R2 - 150-kilo-ohm
- R3-R5 - 1-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 100 μ F, 16V electrolytic
- C3 - 1 μ F, 16V electrolytic
- C4 - 4.7 μ F, 16V electrolytic

Miscellaneous:

- CON2-CON5 - 2-pin connector
- RL1, RL2 - 12V, single-changeover relay
- L1, L2 - 230V AC operated appliance (say 60W bulb)
- X1 - 230V AC primary to 12V-0-12V, 500mA secondary transformer
- Jumper wires
- 16-pin IC socket
- Heat-sink for IC1

low at the same time, which means transistors T2 and T3 are cut-off. This deactivates both the relays and both the appliances are turned off.

Case 5. If you press a key again, the fifth pulse arriving at CLK line again increments the counter by one. Since O_4 is wired to the master reset pin 15 of IC2, it brings CD4017B back to the power-on-reset condition with O_0 high.

The above cycle repeats whenever you press a key on the remote. Thus, the circuit simply operates as a two-channel on/off toggle switch for appliances, which is controlled using an IR remote device.

Construction and testing

An actual-size PCB layout for the two-channel IR remote control switch is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on PCB, connect the secondary of transformer across X1.

In place of 230V AC operated appliances, you may use 60W, 230V AC bulbs for testing, as shown in the prototype (refer Fig. 1). **EFY**



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